

Helix OTA — UI/UX + OpenDesign (§11.4.162) + Host-Render Visual-Proof (§11.4.170) Audit

Date: 2026-07-09 **Trigger:** Operator question — “Does Helix OTA have properly designed UI/UX for all client apps and all frontends on all OSes/platforms, and is it OpenDesign-refined and production-polished?” **Method:** Read-only file evidence (manifest inspection + exhaustive grep). No fabrication (§11.4.6). **Author:** background audit subagent (findings relayed + persisted by conductor; subagent Write was blocked by a plugin policy).

Verdict

Real UI surfaces DO exist — two independent React frontends. OpenDesign (§11.4.162) and host-render visual-proof (§11.4.170) are therefore **active, currently-UNMET obligations — not latent.** The project CLAUDE.md note “§11.4.162 latent until this project ships a UI surface” is **stale**: two UIs have shipped.

Surfaces found (file evidence)

#	Surface	Path	User-facing	Evidence
1	OTA Manager SPA	clients/ota-manager/	YES	React 19 + Vite 6 + React Router v7 + Zustand + shadcn/ui (Radix + Tailwind CSS 4) + TanStack Query. Tauri v2 desktop shell (src-tauri/) + experimental Tauri Android/iOS mobile shell. Modules: auth, layout, dashboard, devices, releases, deployments, groups, audit.

#	Surface	Path	User-facing	Evidence
2	Same SPA embedded in Go server	server/ internal/ api/ embed.go + manager-dist/	YES (2nd delivery path of #1)	//go:embed manager-dist/*, MountManagerUI() serves it at / manager.manager-dist/ is the gitignored build output of #1. Not a separate UI.
3	“Operator Dashboard” MVP	dashboard/	YES — a second, distinct impl	React 18 + Vite, hand-rolled src/ components/ui.tsx primitives (not shadcn/ui), 8 screens (Login/ ArtifactUpload/Releases/ Deployments/Fleet/Overview/ Groups/Audit). Named in root README + design doc docs/ research/main_specs/1.0.0-mvp/ dashboard/dashboard_design.md. Has real Playwright e2e + @axe- core/playwright a11y + captured BROWSER_TEST_EVIDENCE.md (5/5 PASS vs a real Go server).
4	submodules/ ota-android-agent/ android/	—	NO	Manifest sharedUserId="android.uid.system" + 3 permissions, zero <activity>; “system-image, not a Play-Store app.” No res/layout/, no @Composable.
5	submodules/ ota-update-engine-bridge/ android/	—	NO	Bare <manifest/>, “carries no components.”
6	ota-artifact-validator, ota-protocol, ota-rollout-engine, ota-telemetry-schema	—	NO	Only tests + docs; no frontend.

#	Surface	Path	User-facing	Evidence
7	Go server	server/	NO (API only)	No html/template/LoadHTMLGlob, no mounted Swagger-UI; only HTML served is the embedded SPA (#2).
8	CLI/TUI client	—	Not found	No such surface exists.

Count: 2 real user-facing UI codebases (clients/ota-manager, dashboard) + 1 duplicate delivery channel of #1.

OpenDesign (§11.4.162)

`grep -rli "open-design|opendesign|open_design"` hits **only governance files** (where the rule text lives). **Zero hits** in dashboard/, clients/ota-manager/, server/, or any package.json/tailwind.config. OpenDesign is **not installed or referenced** in product code.

Used instead: ota-manager = shadcn/ui + Tailwind CSS 4 with `darkMode: "class"` + HSL CSS-variable tokens (a real light/dark token system, but not OpenDesign's). dashboard = hand-rolled `ui.tsx`, no dark-mode.

Host-render visual-proof (§11.4.170)

`grep -rli "toMatchSnapshot|toHaveScreenshot|storybook|chromatic|percy|roborazzi|paparazzi"` → **zero**. dashboard/e2e/*.spec.ts gives genuine live-app Playwright screenshot/video/axe evidence (satisfies §11.4.153/.158/.159) — but NOT §11.4.170's specific ask (host-side, no-running-app, component-level PNG per screen×state×{light,dark} + golden diff + OCR/vision layout oracle). ota-manager has NO visual/e2e harness (only Vitest/RTL unit tests) — the richer frontend is the less-tested one. **§11.4.170 unmet in both.**

Recency / canonical-status data (conductor)

Surface	Last commit	Commits	Named in README/spec?
clients/ota-manager		11	

Surface	Last commit	Commits	Named in README/spec?
dashboard	2026-06-20 (5b0ac61c Auto-commit)	14	Not prominently; is the one embedded at /manager
	2026-06-20 (a9f732dc window-scoped recordings)		Yes — README:103 “Dashboard: React (login, upload, rollout, fleet health)” + formal dashboard_design.md (2026-06-08)

→ Ambiguous which is canonical. **Operator decision required (§11.4.66/§11.4.122)** before sinking OpenDesign effort into a frontend that might be retired.

Prioritized recommendations

1. **[OPERATOR DECISION]** Resolve the two-frontend duplication: is dashboard/ a retired MVP scaffold superseded by clients/ota-manager, or are both intentionally maintained? (§11.4.122 — no silent removal.)
2. Wire OpenDesign into the surface-of-record: install as a dependency, migrate the token surface onto OpenDesign light+dark tokens from Helix brand assets, extend upstream (§11.4.74) for gaps.
3. Add a host-render visual-proof harness (§11.4.170): component render to PNG per screen×state×{light,dark}, golden image-diff + OCR/vision layout oracle.
4. Repeat 2–3 for the second frontend if both are kept; else mark the retired one Obsolete (→ Fixed.md) (§11.4.90) after explicit operator approval.
5. Do **not** apply OpenDesign/§11.4.170 to ota-android-agent / ota-update-engine-bridge — confirmed UI-less system libraries.

Follow-up governance note

The project CLAUDE.md’s “§11.4.162 latent until this project ships a UI surface” line should be corrected — the surfaces exist; §11.4.162 + §11.4.170 are now active obligations. (Tracked as an adoption item.)